PAPER<i>238: UQFF Vacuum Repulsion Force — Surface-Tension Analogy F</i>vac_rep

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Session: *59 (grokshare8d951e12.txt second-pass — Source10)*

PAPER*238: UQFF Vacuum Repulsion Force — Surface-Tension Analogy F*vac_rep

Author: Daniel T. Murphy **Framework:** UQFF v4.9 (Star-Magic) **Session:** 59 (grok*share*8d951e12.txt second-pass — Source10) **Date:** March 2026 **Classification:** Novel UQFF Force — Velocity-Coupled Vacuum Density Contrast Repulsion **Status:** Proof-Quality Whitepaper **CP3 Class:** `UQFFVacuumRepulsionCalculator`

Abstract

This paper introduces $F_{\rm vac\_rep}$, a novel vacuum repulsion force arising from a local contrast in quantum vacuum energy density against a reference background value. Modelled on the analogy of surface tension (which scales with density-contrast at a phase boundary), $F_{\rm vac\_rep}$ scales linearly with the system's bulk velocity and mass, distinguishing it from the dark energy expansion term $F_{\rm DE} = (\Lambda c^2/3)\cdot r$ which is purely radial and velocity-independent. This is the **third distinct UQFF repulsive force** (after $F_{\rm DE}$ and $F_{\rm rel}$) and the only one that couples to instantaneous velocity.

$$\boxed{F_{\rm vac\_rep} = k_{\rm vac}\;\Delta\rho_{\rm vac}\,M\,v}$$

Example result: $F_{\rm vac\_rep} = 1.23\times10^{45}$ N at generic astrophysical scale.

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1. Formula and Parameters

Symbol	Default	Units	Description
$k_{\rm vac}$	6.67×10^{-11}	$m^3/(kg\cdot s^2)$	Vacuum coupling constant (gravitational analogy)
$\Delta\rho_{\rm vac}$	$\rho_{\rm vac,local} - \rho_{\rm vac,ref}$	J/m^3	Local vs reference vacuum energy density contrast
$\rho_{\rm vac,ref}$	1×10^{-9}	J/m^3	Reference quantum vacuum energy density
M	system mass	kg	
v	bulk velocity	m/s	

$$\Delta\rho_{\rm vac} = \rho_{\rm vac,local} - \rho_{\rm vac,ref}$$

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2. Physical Interpretation

The surface-tension analogy captures the following physics:

In ordinary fluid mechanics, surface tension γ acts at a density-contrast boundary $\Delta\rho$

At the quantum vacuum boundary (e.g., edge of a magnetar magnetosphere or stellar wind termination shock), vacuum energy density transitions from $\rho_{\rm vac,ref}$ (ambient) to $\rho_{\rm vac,local}$ (modified by strong fields)

The resulting repulsion scales as $k_{\rm vac}\,\Delta\rho_{\rm vac}\,M$ — analogous to γA — and is amplified by the bulk velocity of material crossing the boundary

Key distinction from $F_{\rm DE}$:

Property	$F_{\rm DE}$	$F_{\rm vac\_rep}$
Origin	Cosmological Λ	Local vacuum density contrast
Radial dependence	$\propto r$ (grows with distance)	Independent of r (surface effect)
Velocity dependence	None	$\propto v$ (linear)
Physical analogy	Hubble flow expansion	Surface tension at vacuum boundary

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3. Scaling Relations

Mass scaling: $F_{\rm vac\_rep} \propto M$ — heavier system experiences stronger vacuum boundary repulsion

Velocity scaling: $F_{\rm vac\_rep} \propto v$ — outflows and infalling material more strongly repelled; at $v=0$, repulsion vanishes

Vacuum contrast: $F_{\rm vac\_rep} \propto \Delta\rho_{\rm vac}$ — vanishes in uniform vacuum (no boundary effect)

Relative strength vs Newtonian gravity:

$$\frac{F_{\rm vac\_rep}}{F_{\rm grav}} = \frac{k_{\rm vac}\,\Delta\rho_{\rm vac}\,v\,r^2}{G\,M}$$

At $r=10^{14}$ m, $v=10^6$ m/s, $\Delta\rho_{\rm vac}=10^{-12}$ J/m^3: ratio $\sim 10^{18}$ (vacuum repulsion dominates at extreme scales).

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4. Novel Contributions

Velocity-coupled vacuum force — first UQFF repulsive term to scale with v

Surface-tension physical model — quantum vacuum boundary physics, not cosmological Λ

Distinct from DE term — $F_{\rm vac\_rep}$ vanishes at rest; $F_{\rm DE}$ is always present

$k_{\rm vac} = G$ — reuses gravitational constant as coupling, establishing dimensional consistency with Newtonian sector

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5. CP3 Implementation

```
calc = UQFFVacuumRepulsionCalculator()
result = calc.compute({
'M': 2.984e31, # kg (Eta Carinae)
'v': 2e6, # m/s (stellar wind outflow speed)
'rho_vac_local': 1e-9 + 5e-13, # J/m³ (slightly enhanced near stellar magnetosphere)
```

```
'rho_vac_ref': 1e-9, # J/m³
})
# result['F_vac_rep'] — vacuum repulsion force (N)
# result['delta_rho_vac'] — vacuum density contrast (J/m³)
```

References

Murphy, D.T. (2025). *Source10 UQFF Catalogue Module*, `UQFFSource10`, `F_vac_rep` definition

grok*share*8d951e12.txt, Source10 Text Module, lines ~5950–5980

Quantum vacuum energy: Casimir, H.B.G. (1948), Proc. K. Ned. Akad. Wet. 51, 793

DE comparison: PAPER_237 (`F_{\rm DE}` as $\Lambda \cdot c^2/3 \cdot r$ term in `F_{U\_Bi\_i}`)